

Max element

6
1 4 8 7 9 3

9 8 4 ~~1~~
ans
2 ~~1~~
location
3
5

```
int ans = a[0], location = 1;
```

```
for(int i = 1; i < n; i++)  
{  
    if(a[i] < ans)  
    {  
        ans = a[i];  
        location = i + 1;  
    }  
}
```

```
cout << ans << " " << location << endl;
```

6	8						
7	6	8	9	1	3		

8

target

0

count



```
int n;  
cin >> n;
```

```
int target;  
cin >> target;
```

```
int a[n];  
for(int i = 0; i < n; i++)  
{  
    cin >> a[i];  
}
```

```
int count = 0;
```

```
for(int i = 0; i < n; i++)  
{  
    if(a[i] == target)  
    {  
        count++;  
    }  
}
```

```
cout << count << endl;
```

6
↓
1 2 2 8 11 31

6
7 6 7 9 7 3

YES

NO

For an array to be in ascending order,
 $a[i] \geq a[i - 1]$ this must be true for every i .

```
int n;  
cin >> n;  
  
int a[n];  
for(int i = 0; i < n; i++)  
{  
    cin >> a[i];  
}  
  
bool flag = false;  
  
for(int i = 1; i < n; i++)  
{  
    if(a[i] < a[i - 1])  
    {  
        // array is not sorted  
  
        flag = true;  
        break;  
    }  
}  
  
if(flag)  
{  
    cout << "NOT SORTED";  
}  
else  
{  
    cout << "SORTED";  
}
```

10 12 16 17 19

$a[i] < a[i - 1]$

false

flag

6
110110

6
100110

001111

000111

2

c0

4

c1

00000000.....111111111

We can only have 0's and 1's

Print this array in ascending order.

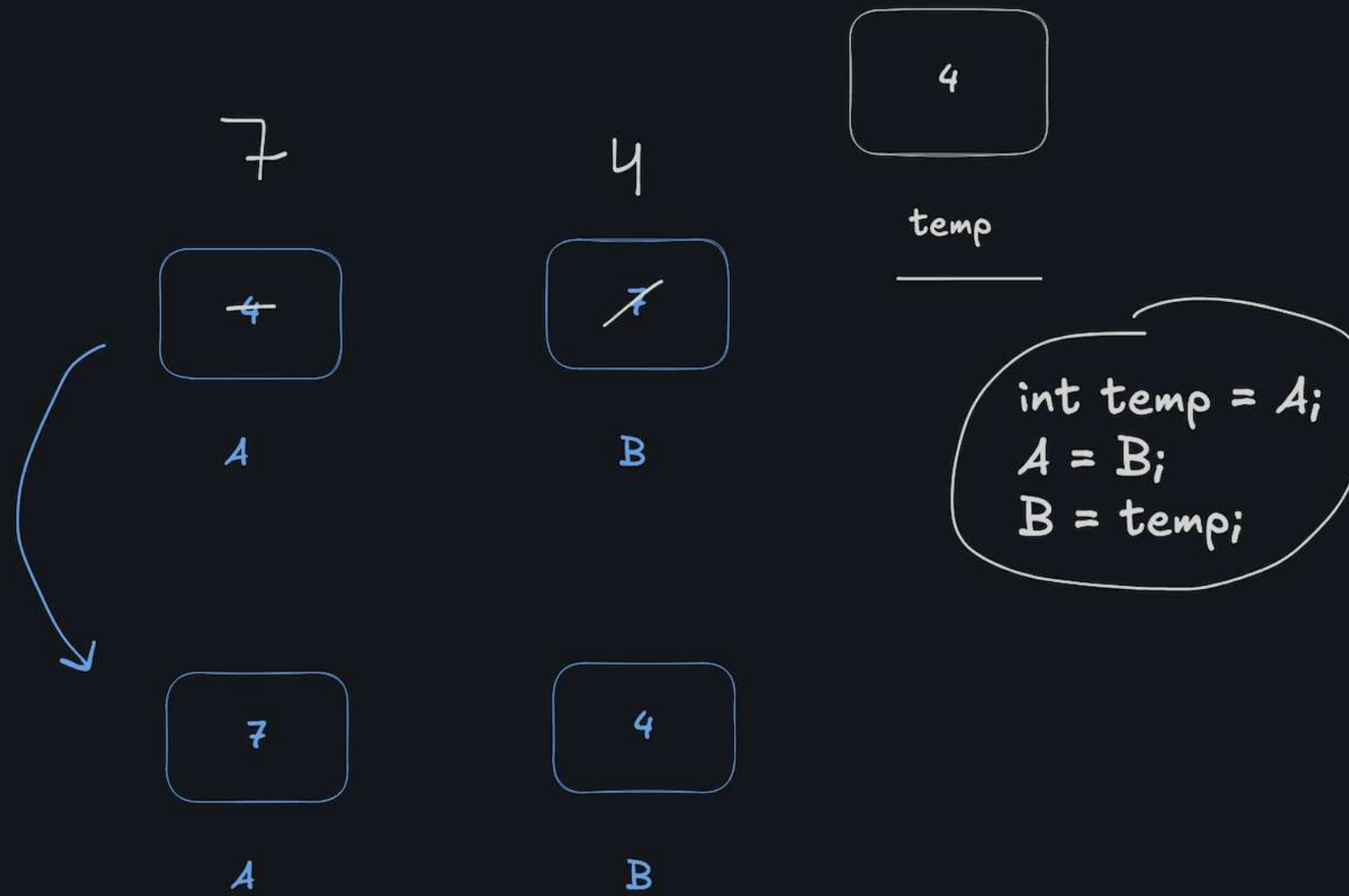
```
int c0 = 0, c1 = 0;

for(int i = 0; i < n; i++)
{
    if(a[i] == 0)
    {
        c0++;
    }
    else
    {
        c1++;
    }
}

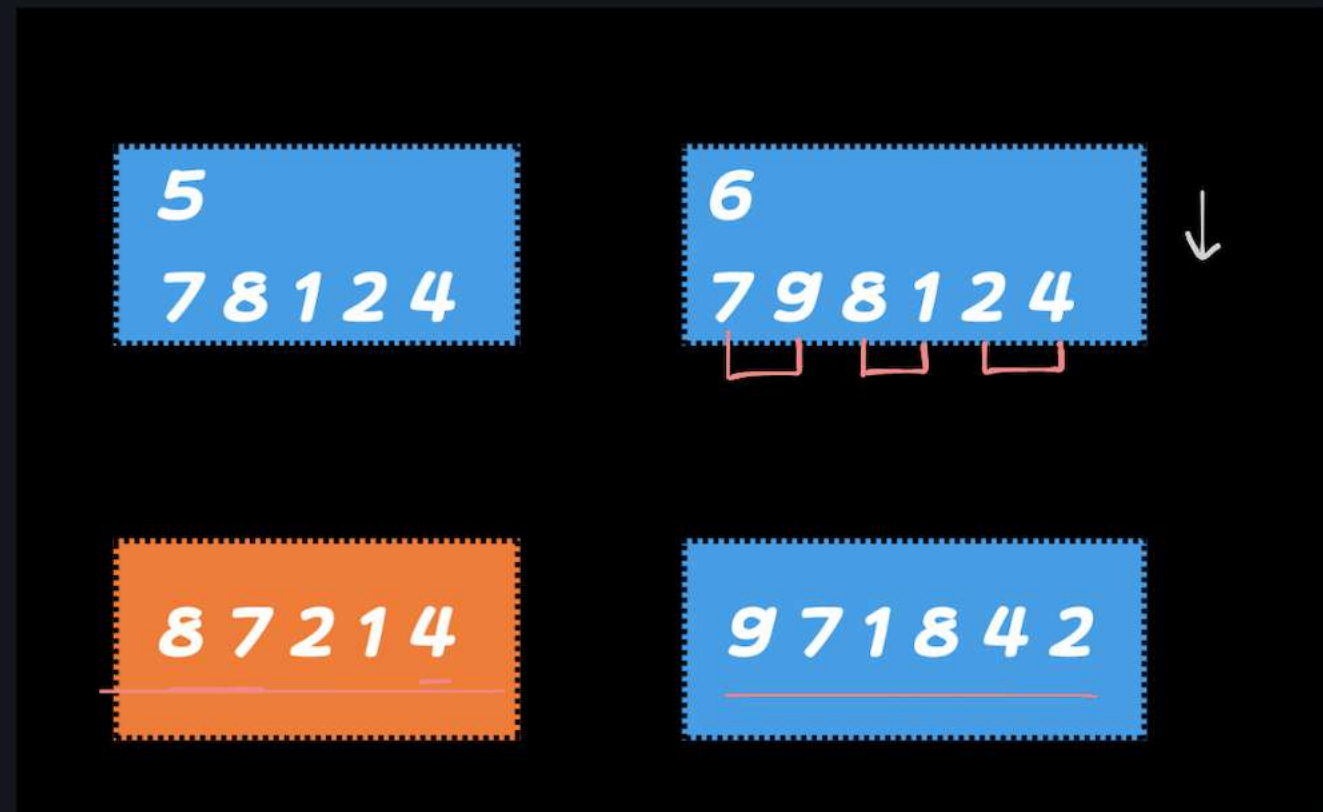
// print all 0's
for(int i = 1; i <= c0; i++)
{
    cout << "0 ";
}

// print all 1's
for(int i = 1; i <= c1; i++)
{
    cout << "1 ";
}
```

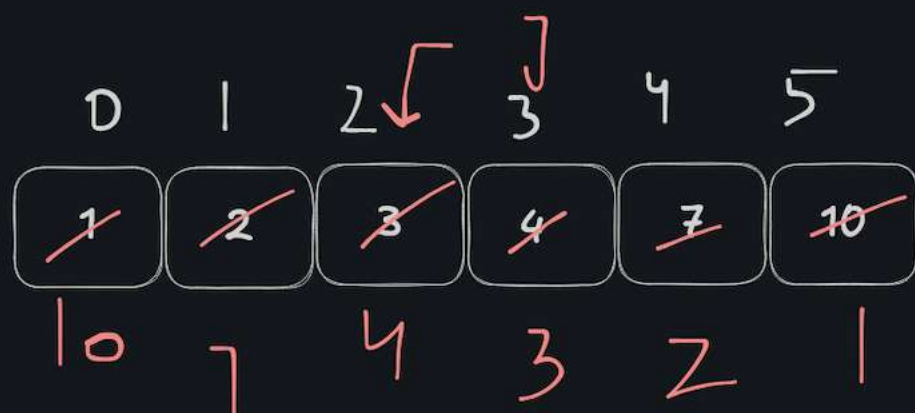
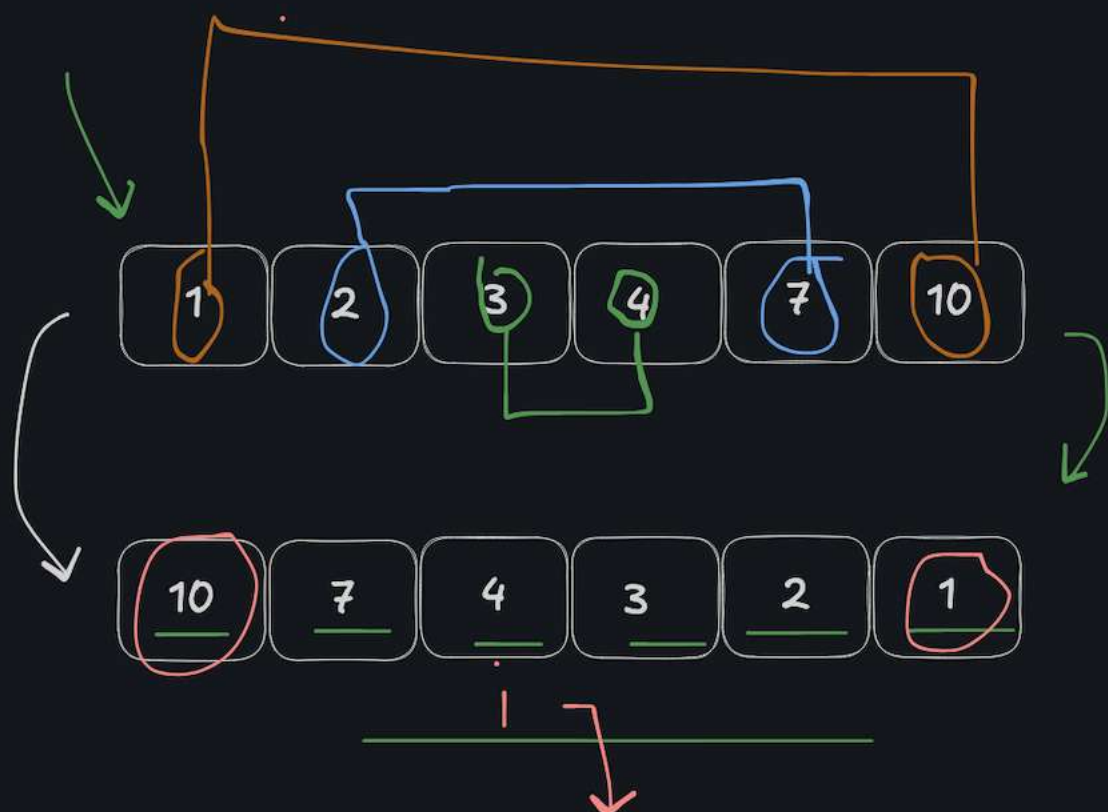
SWAP A and B



Swap alternate elements



```
for(int i = 1; i < n; i += 2)  
{  
    swap(a[i], a[i - 1]);  
}
```

$$i = 3$$

$$j = 2$$

swap

```
int i = 0, j = n - 1;
```

```
while(i <= j)
{
    swap(a[i], a[j]);
    i++;
    j--;
}
```

```
for(int i = 0; i < n; i++)
{
    cout << a[i] << " ";
}
```

MISSING NUMBER

Every number in the array is present twice, except for one special element that occurs only once. Find the special element.



```
int ans;

for(int i = 0; i < n; i++)
{
    // test whether a[i] is the ans or not

    int target = a[i];
    int count = 0;

    for(int j = 0; j < n; j++)
    {
        if(a[j] == target)
        {
            count++;
        }
    }

    if(count == 1)
    {
        ans = a[i];
        break;
    }
}

cout << ans << endl;
```

1 3 5 7 6 4 2

↓ ↓

num 1 3 4 5 6 7 8

```
int ans;
```

```
for(int i = 0; i < n; i++)  
{
```

```
    // check how many times a[i] occurs in the array
```

```
    int count = 0;  
    for(int j = 0; j < n; j++)  
    {  
        if(a[j] == a[i])  
        {  
            count++;  
        }  
    }
```

```
    if(count == 1)  
    {  
        ans = a[i];  
        break;  
    }  
}
```

```
cout << ans << endl;
```


I. Find HCF

time limit per test: 1 second🕒
memory limit per test: 256 megabytes

You are given two integers A and B .

Find the HCF of A and B by writing a function that takes A and B as parameters and returns their HCF.

Input

The only line contains two integers A and B ($1 \leq A, B \leq 10^5$).

Output

Print the HCF of A and B .

Example

input
12 36
output
12

12 -> 1, 2, 3, 4, 6, 12

highest common factor 1, 2, 4

8 -> 1, 2, 4, 8

highest common factor 1, 2, 3, 4, 6, 12

12 -> 1, 2, 3, 4, 6, 12

36 -> 1, 2, 3, 4, 6, 9, 12, 18, 36

```
int n, m;  
cin >> n >> m;
```

```
int ans = 1;
```

```
for(int i = 1; i <= n; i++)  
{  
    if(n % i == 0 and m % i == 0) // i is a factor of both n and m  
    {  
        ans = i;  
    }  
}
```

```
cout << ans << endl;
```